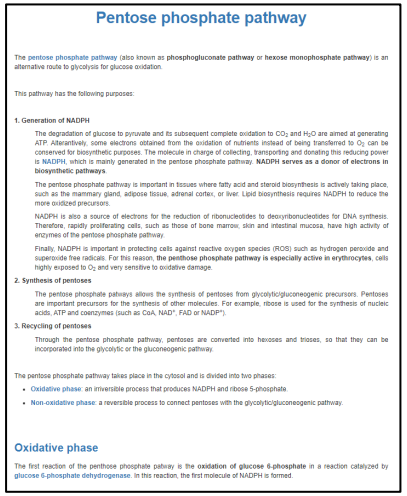


Pentose phosphate pathway

The balance of redox homeostasis

Supportive material



Pentose phosphate pathway

The pentose phosphate pathway (also known as **phosphogluconate pathway** or **hexose monophosphate pathway**) is an alternative route to glycolysis for glucose oxidation.

This pathway has the following purposes:

1. Generation of NADPH

The degradation of glucose to pyruvate and its subsequent complete oxidation to CO_2 and H_2O are aimed at generating ATP. Alternatively, some electrons obtained from the oxidation of nutrients instead of being transferred to O_2 can be conserved for biosynthetic purposes. The molecule in charge of collecting, transporting and donating this reducing power is NADPH, which is mainly generated in the pentose phosphate pathway. **NADPH serves as a donor of electrons in biosynthetic pathways**.

The pentose phosphate pathway is important in tissues where fatty acid and steroid biosynthesis is actively taking place, such as the mammary gland, adipose tissue, adrenal cortex, or liver. Lipid biosynthesis requires NADPH to reduce the more oxidized precursors.

NADPH is also a source of electrons for the reduction of ribonucleotides to deoxyribonucleotides for DNA synthesis. Therefore, rapidly proliferating cells, such as those of bone marrow, skin and intestinal mucosa, have high activity of enzymes of the pentose phosphate pathway.

Finally, NADPH is important in protecting cells against reactive oxygen species (ROS) such as hydrogen peroxide and superoxide free radicals. For this reason, **the penthose phosphate pathway is especially active in erythrocytes**, cells highly exposed to O_2 and very sensitive to oxidative damage.

2. Synthesis of pentoses

The pentose phosphate pathways allows the synthesis of pentoses from glycolytic/gluconeogenic precursors. Pentoses are important precursors for the synthesis of other molecules. For example, ribose is used for the synthesis of nucleic acids, ATP and coenzymes (such as CoA, NAD^+, FAD or $NADP^+$).

3. Recycling of pentoses

Through the pentose phosphate pathway, pentoses are converted into hexoses and trioses, so that they can be incorporated into the glycolytic or the gluconeogenic pathway.

The pentose phosphate pathway takes place in the cytosol and is divided into two phases:

- Oxidative phase: an irriversible process that produces NADPH and ribose 5-phosphate.
- Non-oxidative phase: a reversible process to connect pentoses with the glycolytic/gluconeogenic pathway.

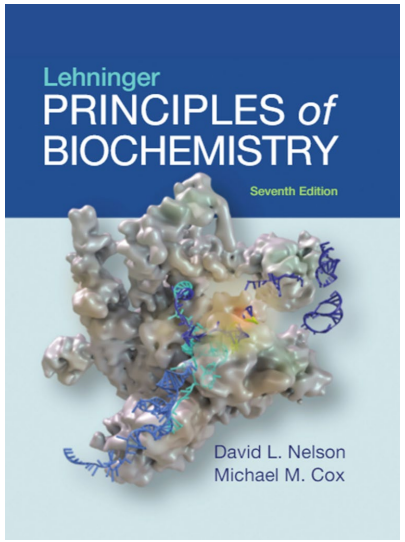
Oxidative phase

The first reaction of the penthose phosphate pathway is the **oxidation of glucose 6-phosphate** in a reaction catalyzed by glucose 6-phosphate dehydrogenase. In this reaction, the first molecule of NADPH is formed.

Additional study material

ADI document:

Pentose phosphate pathway



Reading

Lehninger’s PoB Chapter 14:

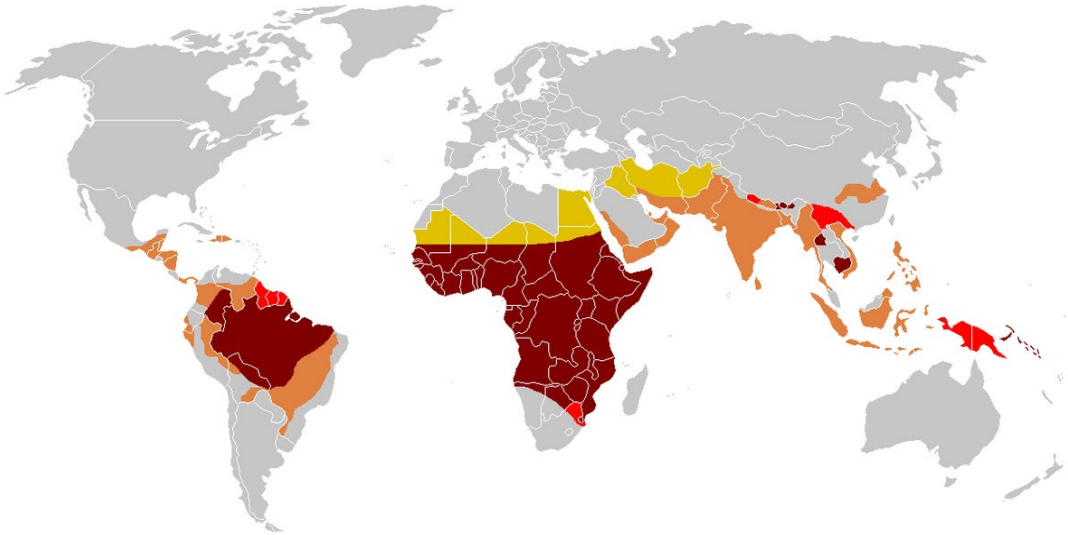
14.5. Pentose phosphate pathway of glucose oxidation

Learning goals

- Objectives, phases and regulation of the pathway
- Glutathione for the control of oxidative stress
- G6PDH deficiency
- Alternative uses of glycolysis and pentose phosphate pathways

Malaria

- Malaria is caused by unicellular parasites of the Plasmodium group
- The disease is widespread in the tropical and subtropical regions around the equator

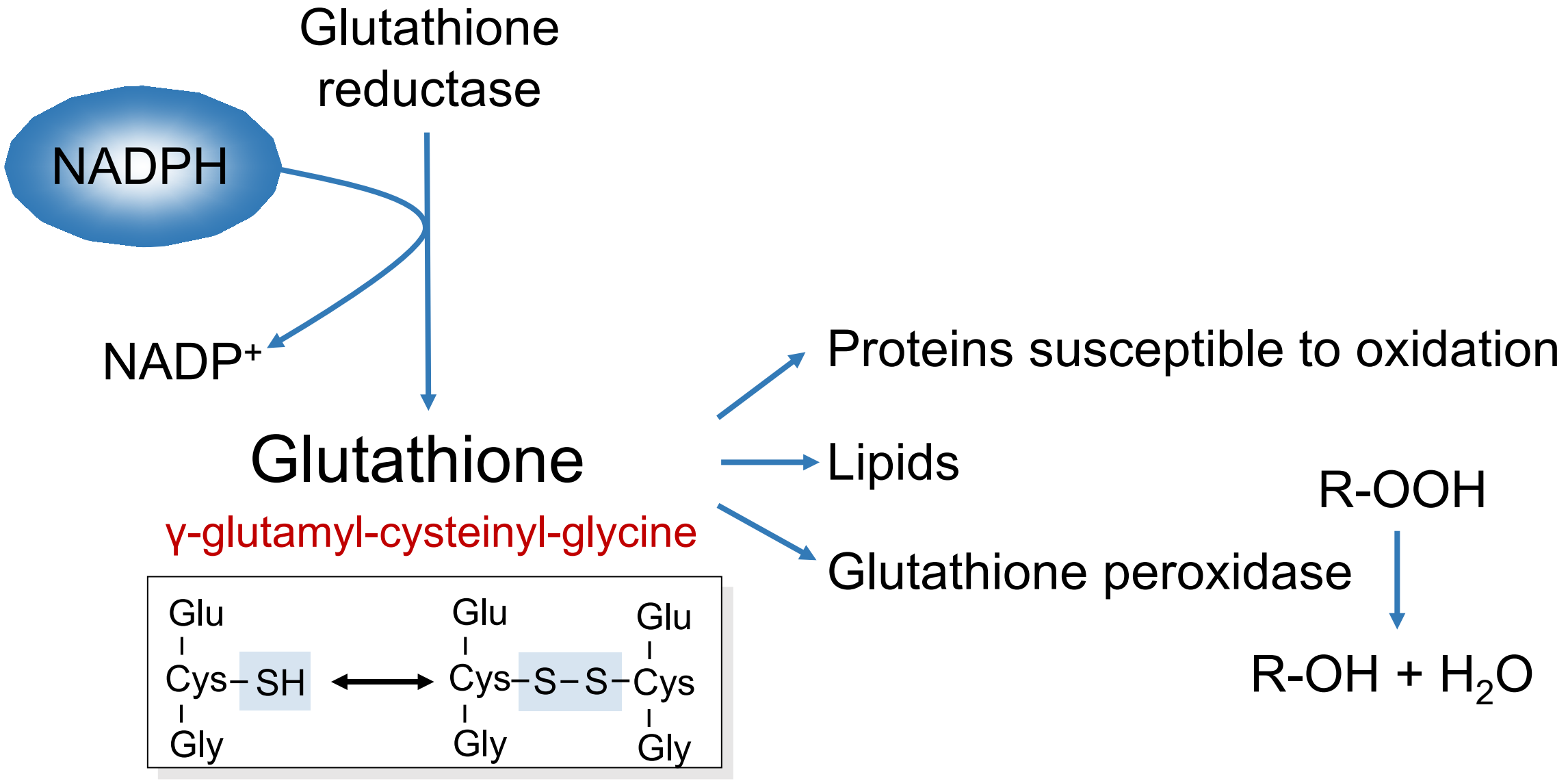


- Symptoms include fever, tiredness, vomiting and headaches. In severe cases, it can cause seizures, coma and death

Oxidative stress and malaria

- Plasmodium parasite development occurs in red blood cells
- Parasites are highly susceptible to oxidative stress in infected erythrocytes
- Therefore, the control of oxidative stress is a critical factor for the malaria parasite

Antioxidant control of oxidative stress



Pentose phosphate pathway: objectives

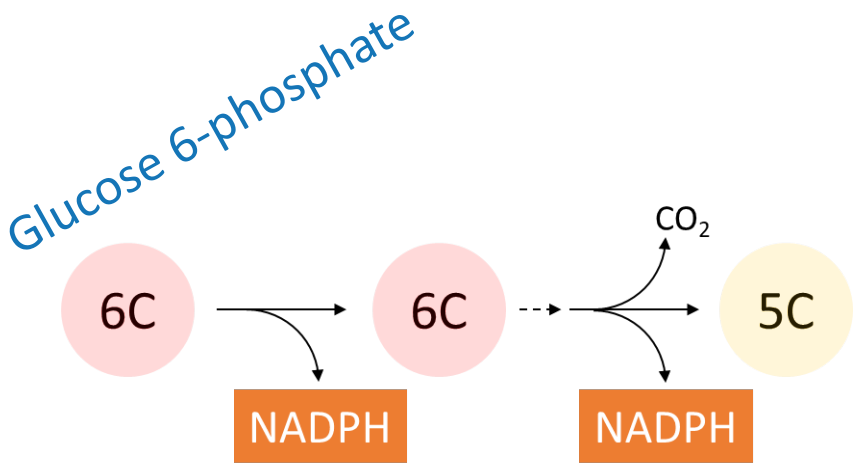
- The principal source of NADPH, an electron donor for:
    - Biosynthesis of lipids or DNA
    - Redox homeostasis and repair of oxidative damage
- Production of pentoses (such as ribose 5-phosphate for the synthesis of nucleic acids and some coenzymes)
- Entry of pentoses into the glycolytic/gluconeogenic pathway

Pentose phosphate pathway: characteristics

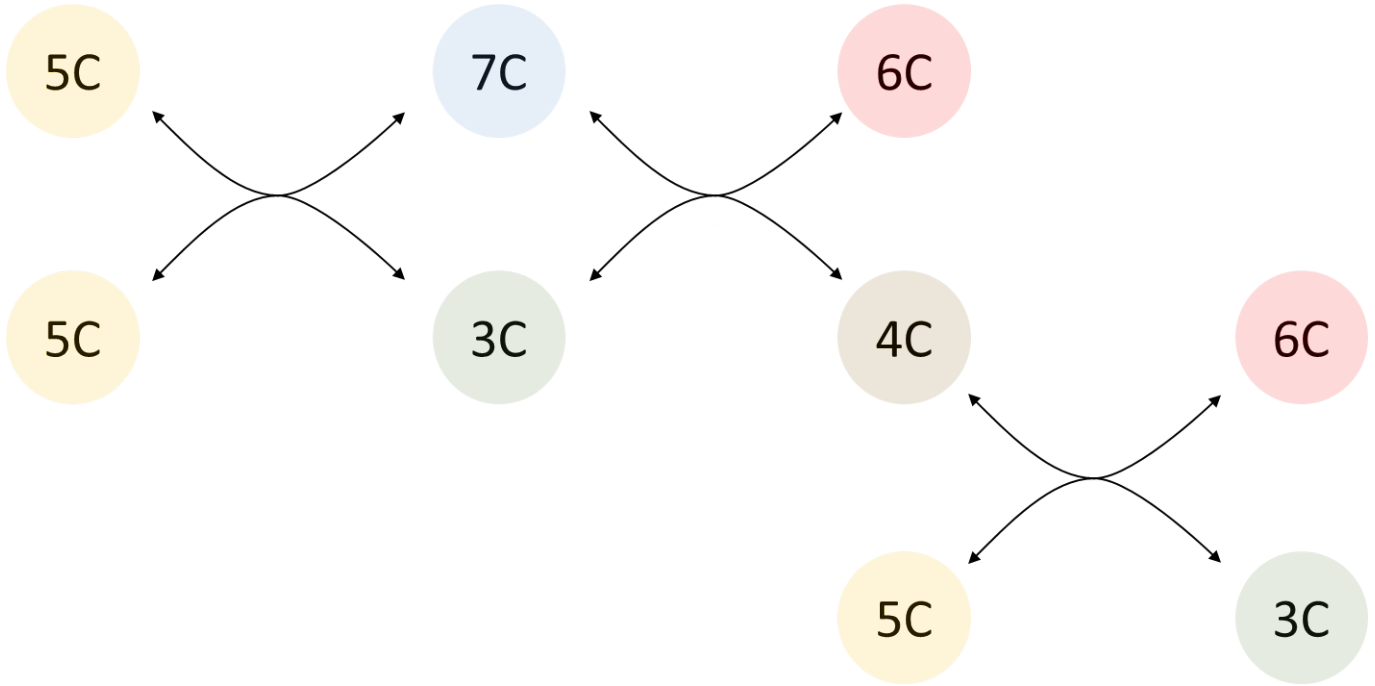
- Cytosolic
- Both anabolic and catabolic
- Two phases:
  - Oxidative phase
    - *Produces NADPH and ribulose 5-phosphate*
    - *Irreversible*
  - Non-oxidative phase
    - *Incorporates pentoses into the glycolytic/gluconeogenic pathway*
    - *Reversible*

Pentose phosphate pathway: overview

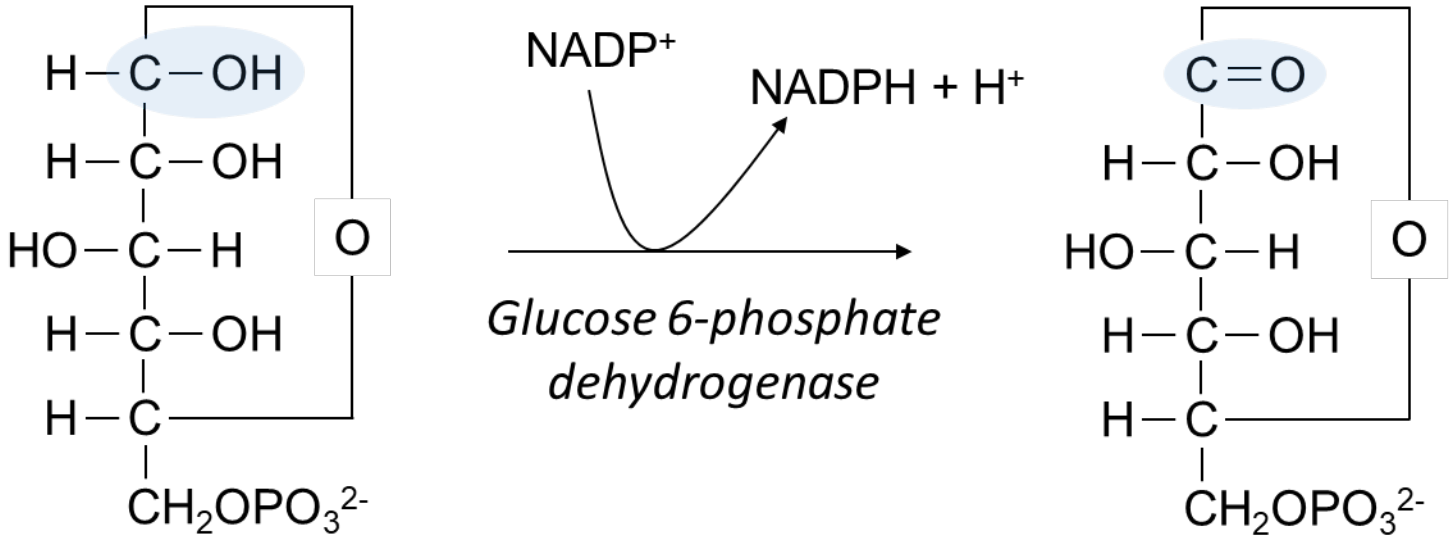
Oxidative phase



Non-oxidative phase



Oxidation of glucose 6-phosphate

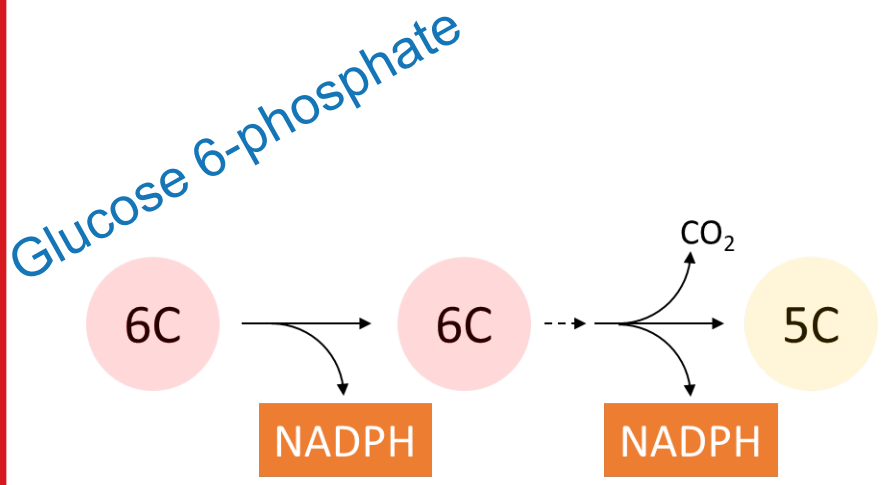


Glucose 6-phosphate

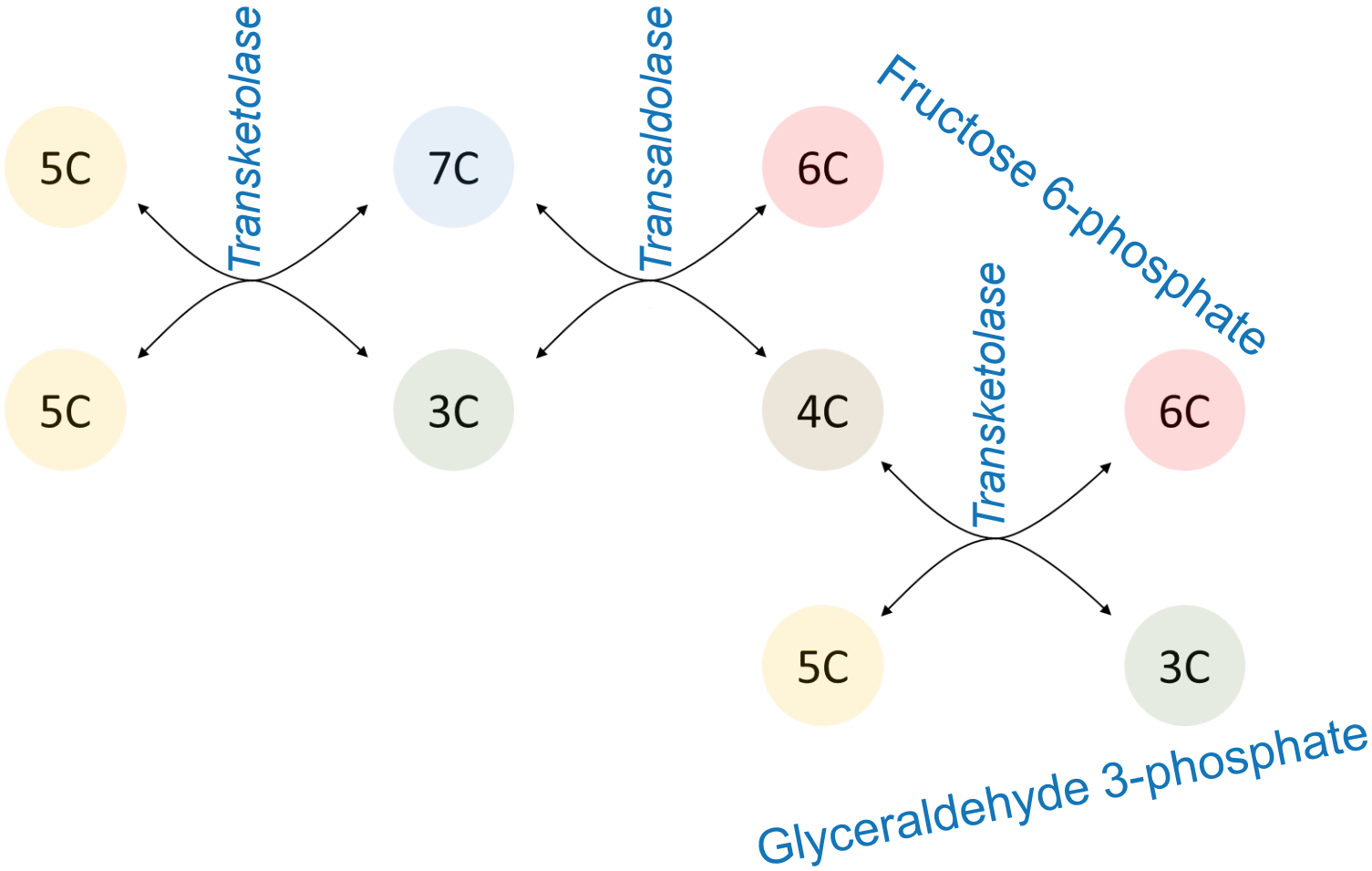
6-Phosphoglucono-δ-lactone

Pentose phosphate pathway: overview

Oxidative phase



Non-oxidative phase



Pentose phosphate pathway: regulation

- Oxidative phase

  Glucose 6-phosphate dehydrogenase ⊖ NADPH

- Non-oxidative phase

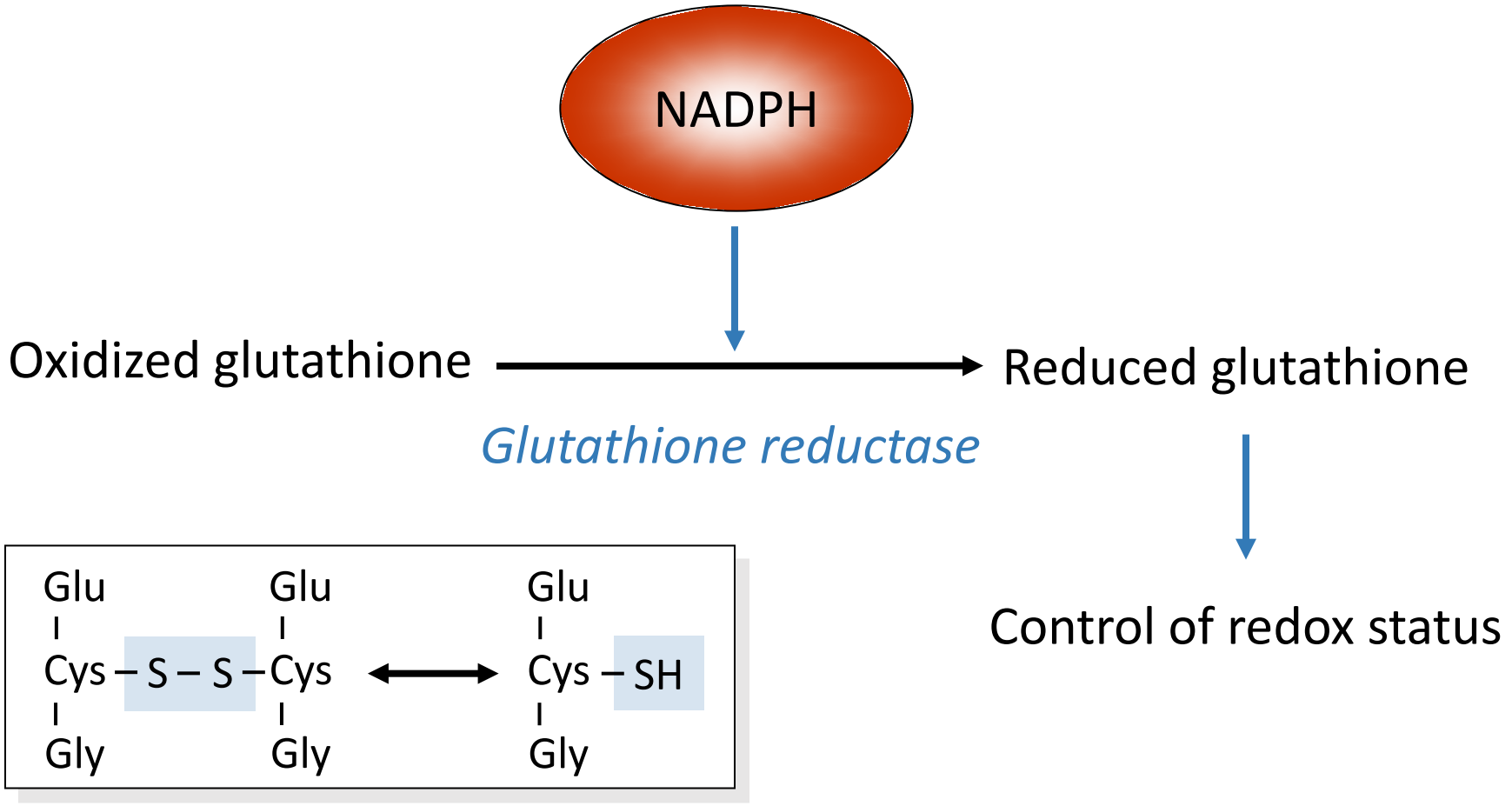
  [Metabolites]

Glucose 6-phosphate dehydrogenase deficiency

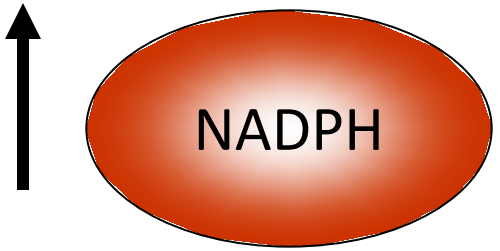
- One of the most common enzyme deficiencies in the world
  - 400 million people affected
- X-linked
- Reduction of enzyme activity
- Most individuals are asymptomatic under normal conditions
- Can be fatal in cases of high oxidative stress
  - Certain drugs, herbicides, and fava beans



- Confers some resistance to malaria due to higher oxidative stress in red blood cells

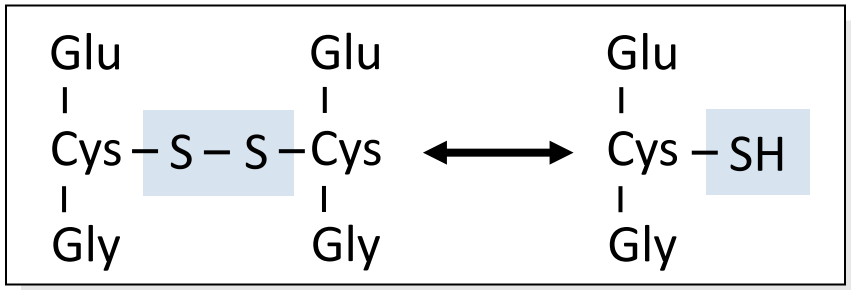

NADPH
Oxidized glutathione
Reduced glutathione
Glutathione reductase
Control of redox status
Glu
Cys – S – S – Cys
Gly
Glu
Cys – SH
Gly

Oxidative damage



Oxidized glutathione → Reduced glutathione

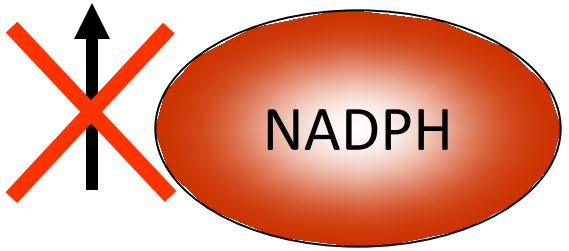
Glutathione reductase



Control of redox status

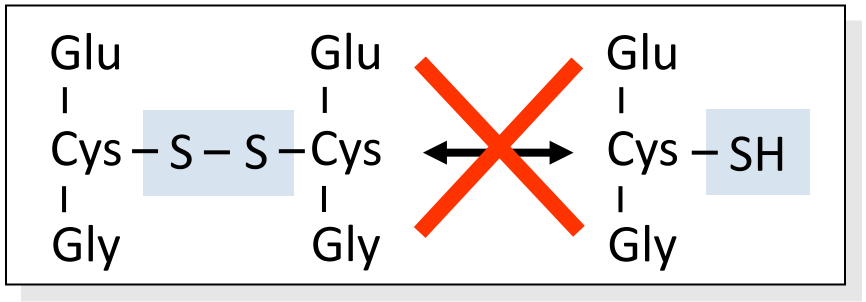
Oxidative damage

Deficit of G6PD

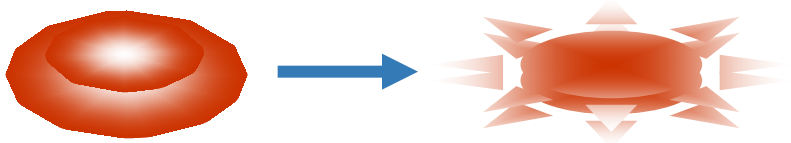


Oxidized glutathione → Reduced glutathione

Glutathione reductase

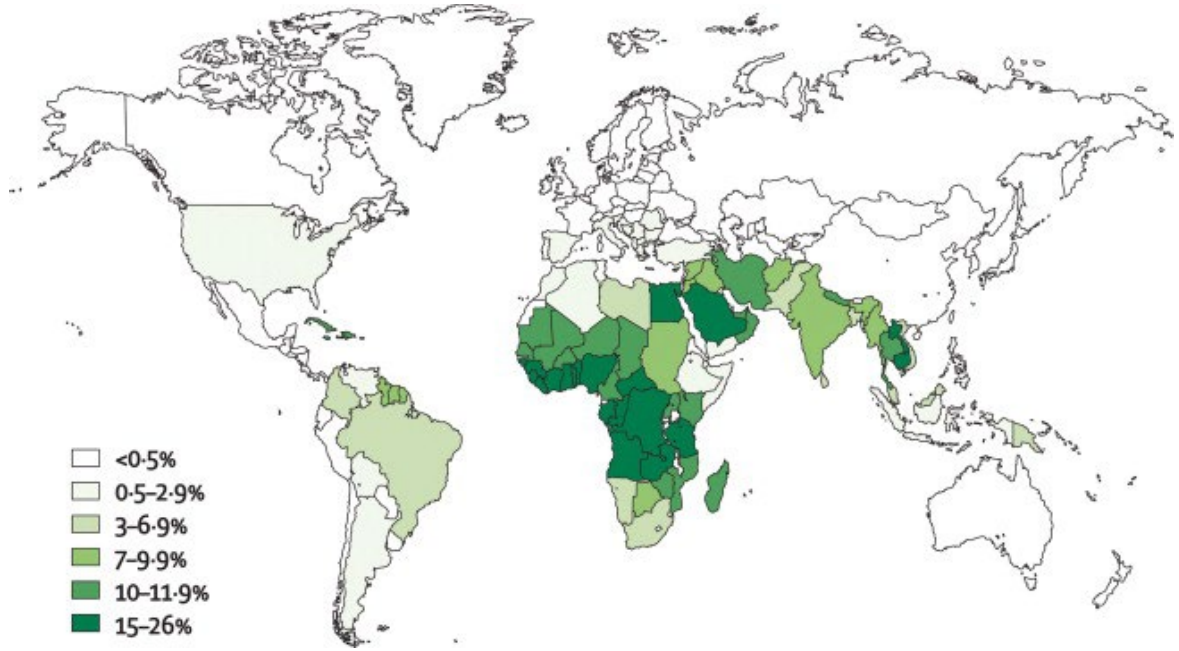


Control of redox status

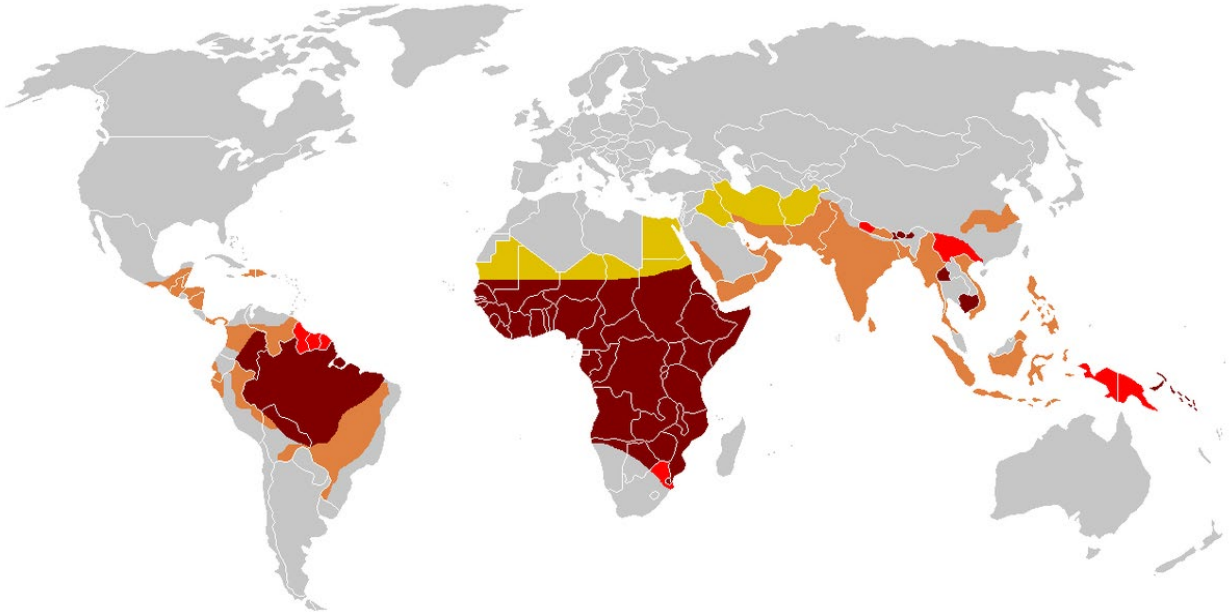


Hemolytic anemia

Geographic distribution of G6PD deficiency



Geographic distribution of malaria

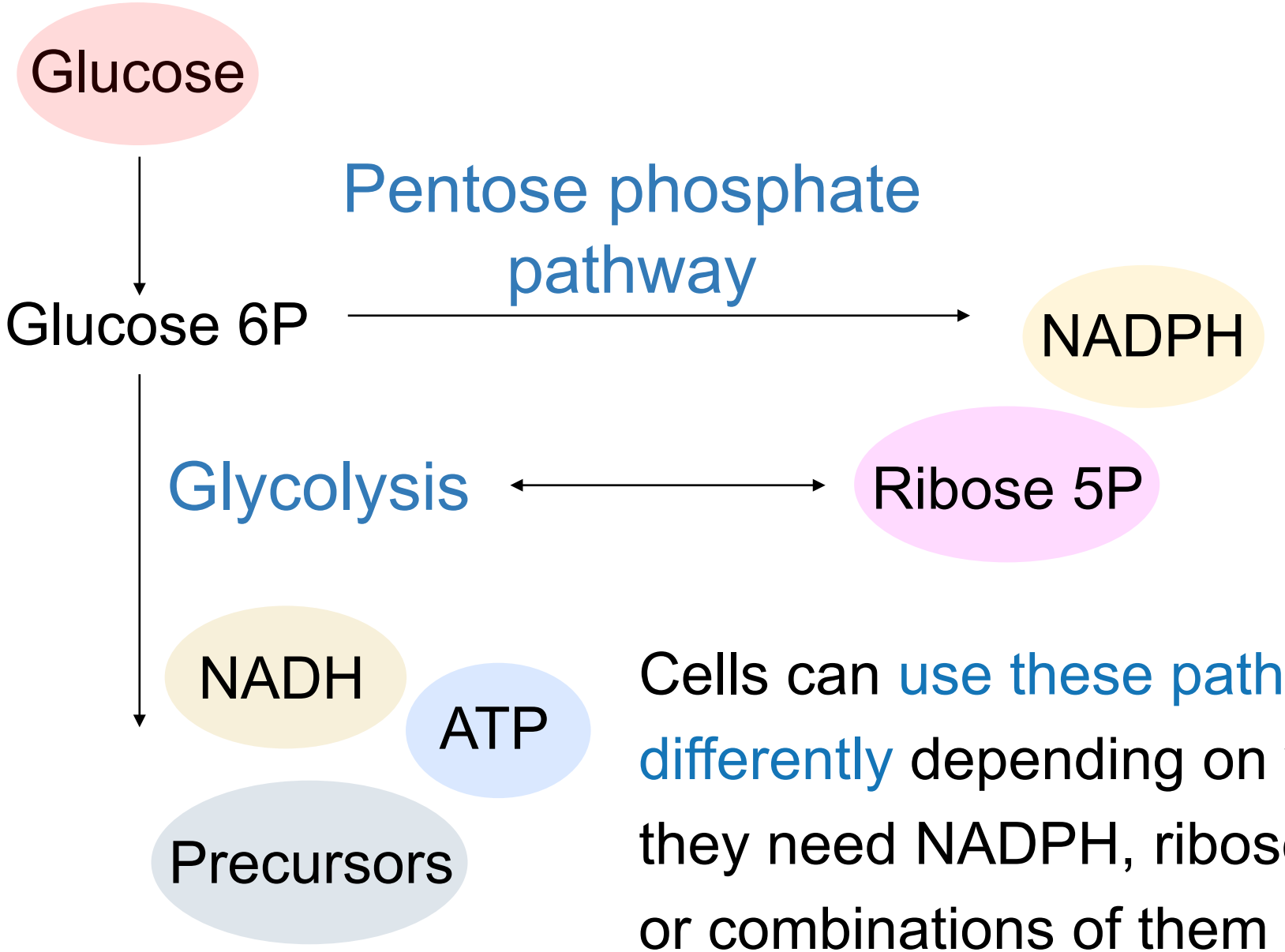


WHO working group. *Bull World Health Organ*.1989; 67: 601–611

Quinine

- The first effective treatment for malaria came from the bark of cinchona tree, which contains quinine
- Quinine interferes with heme detoxification in *Plasmodium*, increasing toxic stress in the parasite
- Paradoxically, quinine treatment in G6PD-deficient individuals, who are more resistant to malaria, can increase oxidative stress and trigger hemolytic anemia

Alternative use of glucose



Cells can use these pathways differently depending on whether they need NADPH, ribose 5P, ATP, or combinations of them

Cell that requires both NADPH and ribose

Cell that requires NADPH and ATP

Cell that requires NADPH